

Optimizing Supply Chain Management of Fresh E-Commerce Agri-Consumer Products Using Energy-Efficient Vehicle Routing

Siraj Zahran 

Abstract—Nowadays, optimizing supply chain management of fresh Agri-products has become a major challenge for food distributors due to the short shelf-life and perishable nature of these products. The prompt and high-quality distribution of these goods without sacrificing their freshness, however, immediately raises the cost of transportation. Considering these facts, the current study attempts to investigate the function of several characteristics that affect the distribution network's efficacy. Afterwards, an optimization model is developed to reduce the vehicle travel-times using an energy-efficient wireless vehicular network. This model ensures that all constraints of the delivery process are met under ambient conditions. The primary goal is to minimize the overall delivery cost without causing freshness decay. To achieve this, an enhanced ant colony algorithm (ACA) is tailored for open vehicle routing instead of closed vehicle routing strategies. The validation of the model is done through comparison of Improved Ant Colony Algorithm (IACA) with Ant Colony Algorithm (ACA). Results indicate that the Improved Ant Colony Algorithm (IACA) is significantly cheaper for the delivery of fresh e-commerce products. Additionally, this approach utilizes an energy-efficient wireless vehicular network, which further enhances the sustainability and efficiency of logistics operations.

Index Terms—E-commerce, supply chain management, perishable products, energy-efficient routing, wireless vehicular network.

I. INTRODUCTION

FRESH e-commerce has emerged as a major channel for the distribution of agricultural products, experiencing substantial growth globally. In recent years, the market transaction volume of the fresh e-commerce industry has seen a dramatic increase, highlighting its potential. The potential of the new e-commerce business has been shown by a sharp rise in market transaction volume in recent years. The perishable nature of fresh agricultural products presents difficulties, though. Consumers frequently make small, frequent purchases, which means that delivery windows must be specified and large expenses associated with transportation and logistics are incurred. Customers often make frequent, small purchases, necessitating specific delivery windows and thus incurring high logistics and delivery costs. Consequently, logistics and

delivery have become vital for the success of fresh agricultural product e-commerce.

The scientific planning of delivery vehicle routes, specifically addressing the Vehicle Routing Problem (VRP) for fresh agricultural product distribution, is crucial. It is imperative to plan delivery vehicle routes scientifically, with a focus on solving the Vehicle Routing Problem (VRP) in the distribution of fresh agricultural products. Most of the research in this field focuses on VRP in normal temperature environments, encompassing a range of topics including reducing delivery costs, handling market risks, and factoring in customer happiness. Notably, existing literature such as [1], [2], [3], [4] provides valuable insights into these aspects.

Cold chain logistics has also been recognized as a critical factor in maintaining the quality of fresh agricultural product distribution, with numerous studies developing models and algorithms for effective cold chain distribution, as detailed in [5], [6], [7], [8], [9]. With relevant findings presented in literature [10] and [11]. A sensor network is a collection of sensors, each of which gathers data from a distinct area and transmits it to a central point for processing, archiving, and viewing. Vehicular networks are a subset of mobile computer networks in which individual automobiles interact with one another to serve as network nodes. Another facet is urban distribution, where research is concentrated on optimizing delivery routes in metropolitan areas with fluctuating traffic circumstances utilizing the Time-Dependent Vehicle Routing Problem (TDVRP), as discussed in references. Urban distribution presents another dimension, with studies focusing on the Time-Dependent Vehicle Routing Problem (TDVRP) to optimize delivery routes in urban areas with variable traffic conditions, as explored in [12], [13], [14], [15].

This paper aims to fill these gaps by developing mathematical models and solution algorithms for the open time-dependent vehicle routing problem (OTDVRP) under freshness constraints, offering new insights for fresh e-commerce logistics and decision-making. This is how the time dependent vehicle routing problem (TDVRP) is defined. A central depot serves customers with fixed requests using a fleet of vehicles with fixed capacities. Vehicles are assigned to customers, and the vehicles are routed to minimize the overall time of the journeys. The perishable and short shelf life of fresh agricultural products have made chain management a significant concern for food distributors. Cold chain distribution using models and algorithms developed by multiple studies,

Manuscript received 25 November 2023; revised 12 January 2024 and 15 February 2024; accepted 24 February 2024. Date of publication 27 February 2024; date of current version 26 April 2024.

The author is with the Industrial Engineering Department, University of Business and Technology, Jeddah 21432, Saudi Arabia (e-mail: s.zahran@ubt.edu.sa).

Digital Object Identifier 10.1109/TCE.2024.3370845

with a focus on fresh agricultural product delivery. There are challenges, though, because fresh agricultural products are perishable, enhanced consumer happiness, lower operating costs, and enhanced supply chain management are all benefits of this in the fresh e-commerce industry.

The primary objective of this work is to develop innovative mathematical models and solution algorithms for the Open Time-Dependent Vehicle Routing Problem (OTDVRP) under freshness constraints. This aims to address specific challenges faced in the distribution of fresh agricultural products through e-commerce channels. Usually obtained directly from farms or agricultural organizations, perishable foods, often known as perishables, are raw or fresh versions of crops. Perishables, which are typically more organic or natural, are transported from farmers to customers with minimum processing or modification. The focus is on optimizing delivery routes in a way that accounts for the perishable nature of the products, the variability of urban traffic, and the diverse needs of customers for fresh products. To increase the fresh e-commerce industry's logistics operations' efficacy and efficiency, which will lower operating costs, boost consumer happiness, and improve supply chain management overall. In the fresh e-commerce sector, thereby contributing to reduced operational costs, improved customer satisfaction, and overall better management of the supply chain. Because of its vertically linked supply chains, the consumer electronics industry faces substantial risks, even while some of the supply- and demand-related issues are shared by other industries, such as the toy sector. In the consumer electronics industry, capacity can easily be too high or too low. The consumer electronics sector is the perfect example of these supply chain management characteristics and associated risks.

The contribution of this work is multifaceted, aimed at advancing the field of logistics in fresh e-commerce through the development of innovative mathematical models and solution algorithms for the Open Time-Dependent Vehicle Routing Problem (OTDVRP) under freshness constraints. It introduces novel mathematical formulations that explicitly incorporate the unique challenges of fresh product distribution, such as perishability, time-dependent urban traffic, and diverse customer delivery requirements. The maintenance of the quality of fresh agricultural product distribution has been found to be significantly influenced by cold chain logistics. Various research has developed models and algorithms for efficient cold chain distribution, as referenced. An increasing emphasis is also being placed on distribution under freshness limits, considering the needs of customers regarding product freshness as well as operational expenses. The work also develops and tests advanced solution algorithms. Creates and evaluates sophisticated algorithms for solving these challenging routing problems, maybe utilizing heuristic and metaheuristic methods. One somewhat neglected component of the routing challenge is the integration of freshness limitations. This research also addresses practical, real-world logistical challenges, providing valuable insights for small and medium-sized e-commerce enterprises that outsource delivery operations. Utilizing a wireless vehicular network that is energy-efficient, the vehicle trip times. Groups of moving or

stationary vehicles connected by a wireless network make up a vehicular ad hoc network (VANET). VANETs were mostly used up until recently to give drivers comfort and safety in moving vehicles. It involves a comprehensive analysis of time-varying urban road networks and their impact on delivery efficiency, an important yet often overlooked factor in urban logistics. Overall, by guaranteeing the prompt delivery of fresh agricultural items, this effort aims to improve operational efficiency, lower costs, and raise customer happiness by providing fresh e-commerce enterprises with improved decision-making tools.

The challenge of finding the most economical delivery routes or directions from a depot to a collection of consumers who are dispersed geographically, with an emphasis on transverse limitations, is known as vehicle routing problem (VRP) [16], [17]. Within the framework of supply chain and logistics management, VRP is in charge of product and service distribution. Behavior monitoring could be completely changed by incorporating artificial intelligence (AI) into human activity recognition (HAR) in smart surveillance systems. These systems improve security and surveillance measures by continuously monitoring and identifying any anomalous or suspicious activity in video feeds by analyzing an individual's physiological or behavioral attributes [18]. Microgrids (MG) are a major help to main grids for efficient energy distribution, serving as a link between power generation and consumption. Although renewable energy resources—photovoltaics in particular—are thought to be clean energy sources, forecasting them can be difficult due to their complicated, volatile, and intermittent nature [19]. The Internet of Things (IoT) and communication both greatly benefit from Human Activity Recognition (HAR), which makes it possible for visual sensors to interpret and react to human behaviour more effectively and intelligently. Current deep learning models need a lot of processing power and are not suitable for edge devices because of their inability to handle low light levels, a variety of views, and cluttered backgrounds [20]. This field's study focuses on variable rate pricing (VRP) in surroundings with typical temperatures and covers a wide range of concerns, such as lowering delivery costs, managing market risks, and accounting for customer satisfaction. To evaluate the viability and efficacy of the IACA algorithm presented in this research, a range of case types were used in the trials.

II. PROBLEM DESCRIPTION

A recently established online retailer contracts with a third-party logistics firm to handle its delivery needs. There are several customers whose delivery windows and locations are known.

The following presumptions are made to define the scope of this paper: (1) Vehicles are not required to return to the e-commerce company's warehouse upon completion of a task; instead, they can leave at any time. (2) Although vehicles are permitted to arrive early at customer locations, deliveries cannot be made until the customer's earliest service start time. The traffic road network displays time-varying elements, and the vehicle's journey speed varies at different periods.

Because of the fixed costs associated with vehicle utilization, labor costs are based on how much time the vehicle is used, including travel, waiting, and maintenance time. (3) When delivered to clients, fresh agricultural products are subject to a minimum freshness constraint. (4) The vehicle's journey speed varies at different times, and the traffic road network exhibits time-varying features. Fresh agricultural products are subject to a minimum freshness constraint when delivered to clients. The government uses the Minimum Support Price (MSP) as a type of intervention to protect farmers against sharp drops in the market pricing of their products and to assist them in avoiding losses. (5) The fixed usage costs of vehicles result in labor costs that are determined by the amount of time the vehicle is used, including travel, waiting, and servicing time.

Develops mathematical models and algorithms to solve the open time-dependent vehicle routing problem in an effort to close these gaps. Additionally, in order to effectively handle these challenging routing problems, the work develops and tests sophisticated solution algorithms, possibly incorporating heuristic and metaheuristic techniques.

III. MATHEMATICAL MODEL

A. Calculation of Travel Time Across Time Periods

Due to reasons such as morning and evening traffic peaks, traffic controls, and unexpected accidents, the traffic conditions of urban road networks have time-varying characteristics, with vehicles traveling at different speeds at different times. Metropolitan road networks experience time-varying traffic conditions, with cars moving at various speeds during different periods of the day. Low power consumption, self-organization, scalability, data aggregation, and strategically positioned sensor nodes are the essential elements of WSNs that allow them to revolutionize data collection and processing for a range of applications. In order to transmit and receive environmental data, a vast number of sensor nodes that are wirelessly connected and densely distributed make up wireless sensor networks, or WSNs. Under time-varying road network conditions, vehicles may need to cross multiple time periods to complete a road segment, making it difficult to calculate travel time directly. One may think of WSN as a subset of IoT. WSN stands for wireless sensor network; IoT is for Internet of items, which combines technologies such as big data analytics, cloud computing/storage, artificial intelligence/machine learning, and networking of items through WSNs. This requires reasonable handling [21]. Endeavours to bridge these gaps by creating mathematical models and algorithms to address the open time-dependent vehicle routing problem. Additionally, in order to properly address these hard routing problems, the work develops and tests advanced solution algorithms, possibly including heuristic and metaheuristic methodologies.

References [22], [23] use the “first-come, first-served” principle to design a method for calculating vehicle travel times under time-varying road network conditions. This paper draws on existing methods and divides the entire 24-hour day into multiple time periods, setting different vehicle travel speeds within each time. Food distributors since these goods

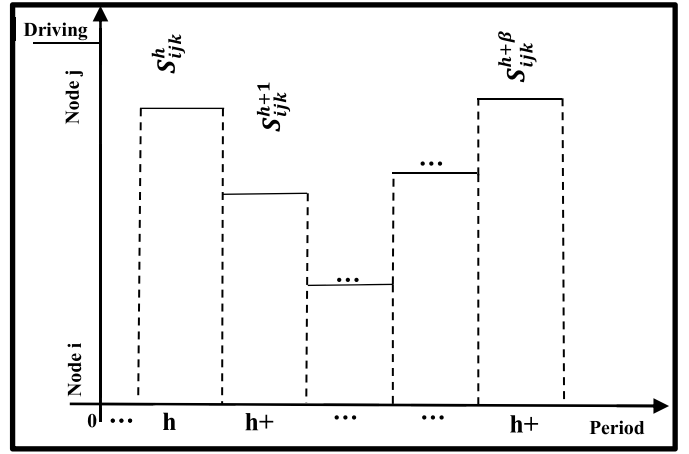


Fig. 1. Vehicle driving speed in each time-period.

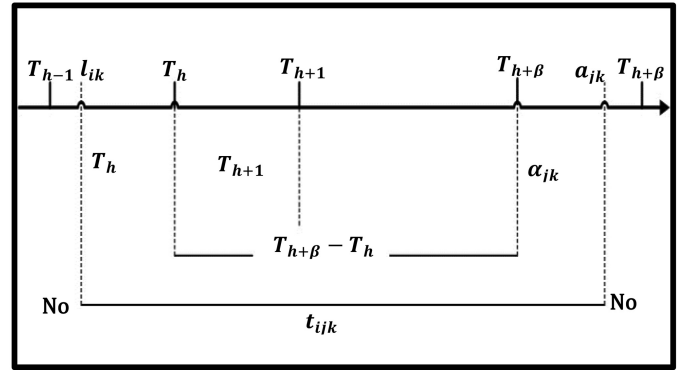


Fig. 2. Vehicle driving time in each time-period.

have a limited shelf life and are perishable. The timely and superior delivery of these products while maintaining their freshness, however, instantly drives up transportation costs. In light of these facts, the current study aims to explore the role of many attributes that influence the effectiveness of the distribution network. Food products classified as perishable must be eaten soon to avoid spoiling. Non-perishable food goods, on the other hand, keep well for a long time. Fresh fruits and vegetables are examples of perishable foods. Foods in cans and dried meats, nuts, and dried fruit are examples of non-perishable foods. Let Z be the fixed time length of a time period; $T = \{T_0, T_1, \dots, T_L\}$ is the set of time periods; $[T_{h-1}, T_h]$ is the h -th time period; S_{ijk}^h represents the travel speed of vehicle k on road segment (i, j) within time period h . The travel speeds of vehicles in different time periods are shown in Figure 1.

Let T_{ijk}^h represent the travel time of vehicle k on the road segment (i, j) within time period h ; l_{jk} is the departure time of vehicle k from node i , and $l_{ik} \in [T_{h-1}, T_h]$; a_{jk} is the time at which vehicle k arrives at node j . According to Figure 2, the travel time of vehicle k on road segment (i, j) within each time period can be divided into three cases: (1) The travel time of the vehicle within the departure time period is $T_h - l_{ik}$. (2) The travel time of the vehicle within the middle time period is Z . (3) The travel time of the vehicle within the last time period is $a_{jk} - T_{h+\beta}$.

Let L_{ij} be the distance on the road segment (i, j); D_{ijk}^h is the travel distance of vehicle k on the road segment (i, j) within time period h, $D_{ijk}^h = S_{ijk}^h T_{ijk}^h$; L_{ij}^h is the distance that vehicle k still needs to travel on the road segment (i, j) after time period h; t_{ijk} is the total travel time for vehicle k to complete the road segment (i, j). The method for calculating travel time across time periods is as follows:

Step 1: Calculation of travel time within the departure time period.

$D_{ijk}^h = S_{ijk}^h (T_h - l_{ik})$. If $D_{ijk}^h \geq L_{ij}$, $t_{ijk} = L_{ij}/S_{ijk}^h$, and the road segment's travel time is calculated. Otherwise, $L_{ij}^h = L_{ij} - D_{ijk}^h$, $T_{ijk}^h = T_h - l_{ik}$, and proceed to Step 2.

Step 2: Calculation of travel time in subsequent time periods.

(1) $\lambda = 1$.

(2) $D_{ijk}^{h+\lambda} = S_{ijk}^{h+\lambda} Z$. If $D_{ijk}^{h+\lambda} \geq L_{ij}^{h+\lambda-1}$ then $T_{ijk}^{h+\lambda} \geq L_{ij}^{h+\lambda-1}/S_{ijk}^{h+\lambda}$, and $t_{ijk} = \sum_{h \in T} T_{ijk}^h$. The road segment's travel time is calculated. Otherwise, $L_{ij}^{h+\lambda} = L_{ij}^{h+\lambda-1} - D_{ijk}^{h+\lambda}$, $T_{ijk}^{h+\lambda} = Z$, $\lambda = \lambda + 1$, and return to (1).

B. Measurement of Freshness for Fresh Agricultural Products

Reference [3] provides a method to calculate the freshness loss of fresh agricultural products over time. A technique for estimating how fresh agricultural products lose their freshness over time. Explains how to gauge how fresh agricultural products are when they are delivered to consumers at room temperature. Reference [11] presents a method for measuring the freshness of agricultural products when they reach customers under normal temperature conditions. Considering that this paper assumes fresh e-commerce typically uses regular vehicles for product delivery, it adopts the freshness measurement function from [11]:

$$X(t_{ik}) = 1 - (t_{ik}/T_b)^2 \quad (1)$$

In the formula, t_{ik} represents the time when fresh agricultural products are delivered to customer i by vehicle k, 1 represents the freshness of fresh agricultural products when they are taken out from the warehouse, T_b is the maximum shelf life of fresh agricultural products, and t_{ik} is less than or equal to T_b .

C. Mathematical Model for OTDVRP With Freshness Constraints

The mathematical model for the Open Time-Dependent Vehicle Routing Problem (OTDVRP) with the objective of minimizing the total delivery cost and under freshness constraints in the context of fresh e-commerce distribution is as follows:

$$\begin{aligned} \min \quad & c_1 \sum_{k \in V} v_k + c_2 \left(\sum_{i \in N} \sum_{j \in C} \sum_{k \in V} \sum_{h \in T} y_{ijk}^h T_{ijk}^h \right. \\ & \left. + \sum_{i \in C} \sum_{k \in V} z_{ik} w_{ik} + \sum_{i \in C} s_i \right) \end{aligned} \quad (2)$$

$$s.t. \quad x_{0jk} = v_k \quad (3)$$

$$\sum_{j \in C} x_{0jk} \leq 1, \forall k \in V \quad (4)$$

$$\sum_{k \in V} z_{ik} = 1, \forall i \in C \quad (5)$$

$$\sum_{i \in N} \sum_{j \in C} x_{ijk} d_i \leq Q, \forall k \in V \quad (6)$$

$$L_{ij} = \sum_{k \in V} \sum_{h \in T} y_{ijk}^h D_{ijk}^h, \forall i \in N, j \in C \quad (7)$$

$$t_{ijk} = \sum_{k \in V} \sum_{h \in T} y_{ijk}^h T_{ijk}^h, \forall i \in N, j \in C \quad (8)$$

$$a_{ik} + w_{ik} \geq b_i, \forall i \in C, k \in V \quad (9)$$

$$a_{ik} + w_{ik} \leq e_i, \forall i \in C, k \in V \quad (10)$$

$$a_{ik} + w_{ik} + s_i = l_{ik}, \forall i \in C, k \in V \quad (11)$$

$$a_{jk} = l_{ik} + t_{ijk}, \forall i \in N, j \in C, k \in V \quad (12)$$

$$t_{ik} = a_{ik} + w_{ik}, \forall i \in C, k \in V \quad (13)$$

$$X(t_{ik}) \geq B \quad (14)$$

$$v_k \in \{0, 1\}, x_{ijk} \in \{0, 1\}, y_{ijk}^h \in \{0, 1\}, z_{ik} \in \{0, 1\} \quad (15)$$

The objective function is represented by equation (2), which minimizes both the labor costs related to vehicle usage time (such as vehicle travel, waiting, and service times) and the fixed costs of vehicle usage. When a vehicle is dispatched, according to constraint (3), it has to leave the warehouse. According to constraint (4), a vehicle can only be utilized once. Only one vehicle may service each customer, thanks to constraint (5). The vehicle load cannot be greater than the vehicle's capacity, according to constraint (6). The link between the length of a road segment and the distance a vehicle goes at various time intervals is explained by constraint (7). Direct trip time calculation is challenging in time-varying road network settings because vehicles may have to cross several time periods to finish a road section. The relationship between a road segment's length and the distance a car travels during different periods of time.

The relationship between a vehicle's travel time on a particular road section and its journey time over various time periods is described in Constraint (8). Constraints show that customer service needs to happen within allotted time slots. The correlation between the time the car arrives at the customer location, waits, is serviced, and leaves. The relationship between the vehicle's arrival time, waiting time, servicing time, and departure time at the customer location is represented by constraint (11). Constraint describes how to calculate the time it takes an automobile to go between nodes. Constraint describes the accurate time computation technique for the delivery of the automobile to a customer. The process for calculating how long it takes a car to go from one node to the next is outlined in Constraint (12). The precise time computation technique for the vehicle's delivery to a customer is described in Constraint (13). Limitation (14) guarantees that fresh agricultural goods are delivered to clients with a minimal level of freshness. Constraint (15) denotes limitations on the values of variables.

IV. SOLUTION ALGORITHM DESIGN

Since VRP is NP-hard, it is challenging to identify an ideal solution. Heuristic algorithms are typically employed to locate workable solutions. Because OTDVRP is more complicated than regular VRP, fixing it is significantly more difficult. This study presents the creation of an Improved Ant Colony Algorithm (IACA) using the following concepts, based on the features of the OTDVRP model: (1) Add a time window width heuristic factor to the transition rules so that clients with small time windows are given priority. (2) Add anticipated and adaptive pheromone heuristic elements to the Ant Colony Algorithm to improve its global search capabilities and keep it from being overly entrenched in local optima too soon. The specific steps of IACA are as follows:

Step 1: Initialization. Initialize all program parameters and variables, set max iter as the maximum number of iterations, iter as the current iteration count, mincost as the best cost, $load_k$ as the current vehicle load, M as the number of ants, $iter = 1$, and $mincost = +\infty$.

Step 2: Construct Feasible Solutions. (1) Place all ants at the warehouse, set $m = 1$. (2) Send out ant m one by one, set $k = 1$. (3) Set $load_{mk}$ to represent the current load of goods for vehicle k used by ant m , $load_{mk} = Q$.

(4) Calculate the transfer probability p_{ij} for ant m 's vehicle k to travel from the current node i to node j :

$$p_{ij} = \begin{cases} \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\delta [\varphi_{ij}]^\vartheta}{\sum_{r \in \text{allowed}_m} [\tau_{ir}]^\alpha [\eta_{ir}]^\delta [\varphi_{ir}]^\vartheta}, & j \in \text{allowed}_m \\ 0, & \text{else} \end{cases} \quad (16)$$

where τ_{ij} is the pheromone concentration heuristic factor, η_{ij} is the visibility heuristic factor, where $\eta_{ij} = 1/L_{ij}$; φ_{ij} is the time window width heuristic factor for node j , $\varphi_{ij} = 1/(e_j - b_j)$; α , δ , and ϑ represent the importance of the pheromone heuristic factor, expected heuristic factor, and time window width heuristic factor, allowed_m represents the set of all unvisited nodes for ant m .

(5) Find the node j_b with the highest transfer probability p_{ij} value and use formula (6) to check whether node j_b satisfies the vehicle capacity limit. Calculate the travel time t_{ijk} for vehicle k from node i to node j_b using the method for cross-period road travel time and use formula (8). Calculate the time t_{ijk} for agricultural products to be delivered by vehicle k to the customer using formula (13). Calculate the freshness $X(t_{ijk})$ of fresh agricultural products delivered to the customer using formula (1). If it meets the constraints of vehicle load capacity, customer time windows, and minimum freshness when delivering fresh agricultural products to customers, vehicle k travels from node i to node j_b , $load_{mk} = load_{mk} - d_{j_b}$, $j_b \notin \text{allowed}$. Otherwise, $k = k + 1$, and send out a vehicle from the warehouse again, returning to (3). (6) If ant m still has unvisited nodes, go back to (4). (7) If all ants have finished visiting, go to step 3; otherwise, $m = m + 1$ and return to (2).

Step 3: Calculate the current iteration results. (1) Calculate the total cost $cost_{iter}$ for the current iteration. If $cost_{iter} < mincost$, set $mincost = cost_{iter}$. (2) Calculate the departure time, vehicle usage time, and travel distance for each vehicle. $iter = iter + 1$.

Step 4: Design adaptive pheromone and expected heuristic factors. Following the method in [24], [25], [26], set $\alpha = 1 + 3(\text{iter}/\text{max iter})$, $\delta = 3 - 2(\text{iter}/\text{max iter})$.

Step 5: Pheromone Update. Update the pheromone levels of the composed road segments of the current iteration's best path, i.e.,

$$\tau_{ij}^{iter+1} = \tau_{ij}^{iter}(1 - \rho) + \sum_{m=1}^M \Delta\tau_{ij}^m \quad (17)$$

$$\Delta\tau_{ij}^m = \begin{cases} F/f_m, \\ \&0, \end{cases} \quad (18)$$

where ρ is the pheromone evaporation rate, $0 \leq \rho < 1$; $\Delta\tau_{ij}^m$ is the increase in pheromone on road segment (i, j) by ant m ; F is a constant; f_m represents the total delivery cost for ant m .

Step 6: Algorithm Termination Condition. If $iter \leq \text{max iter}$, go back to step 2; otherwise, the algorithm ends.

V. CASE ANALYSIS

A. Experimental Setup

Due to the absence of a standard database for fresh agricultural product delivery and the diversity of distribution of fresh e-commerce customers, the C, R, and RC types of VRPTW databases [27] were used as test cases in this study. Qualitative standards for fresh fruit have traditionally included features like appearance, texture, flavour, and nutritional content. However, as the supply chain moves from farm to consumer, safety and traceability become more crucial for all parties involved.

To promote a harmonic balance between effective logistics and environmental conservation by coordinating operational efficiencies in new e-commerce logistics with environmental sustainability goals.

Following the methods in [21], [22], [23], the algorithm parameters were set as follows: $c1 = 200$ CNY/vehicle, $c2 = 0.3$ CNY/min, $Q = 1,000$ units, $\text{max iter} = 600$, $M = 30$, $\rho = 0.15$, $\vartheta = 3$, $F = 20$, $T_b = 780$ min, $B = 0.7$. The algorithm was implemented in MATLAB R2016a and executed on a computer with an Intel® Core™ i5-8250U CPU @ 1.60 GHz 1.80 GHz, and 8.00 GB of RAM.

B. Experimental Results

1. Open Vehicle Routing Analysis for Multiple Case Types

To assess the feasibility and effectiveness of the IACA algorithm proposed in this study, experiments were conducted using various case types with C, R, and RC customer distributions. The computed results are shown in Table I.

Based on the data in Table I, the following conclusions can be drawn:

(a) Based on the calculations of TC, VUT, and VN, it is evident that, relative to the R and RC case types, the C case type has a higher total delivery cost, longer vehicle usage time, and a larger number of vehicles used. The primary reason for this is that the C case type represents a centralized distribution, where each customer is located far from the warehouse (see Figure 3(a) for case C109's delivery route planning). Each vehicle initially delivers to customers near the warehouse and

TABLE I
CALCULATION RESULTS OF DIFFERENT TYPES
OF CALCULATION EXAMPLES

IN	TC/dollar	VUT/min	VTD/km	VN/vehicle	RT/s
C107	5132.8	5773.7	1392.1	18	265.94
C108	3867	4220.9	1158.8	14	260.01
C109	3522.2	3738.3	885.33	13	249.46
R209	3331.9	3770.5	1065.9	12	257.29
R210	3030.2	3431.7	1228.7	11	256.25
R211	2654	2844.3	960.04	10	248.02
RC206	3796.1	4651.2	1244.5	13	263.71
RC207	3479.4	4262.5	1177.2	12	255.51
RC208	2407.5	2689.2	962.37	9	241.57

then to customers closer to the previous customer, reducing the time it takes for the vehicle to reach each customer and improving the freshness of the delivered agricultural products (see Figure 3(b) for case R209 and Figure 3(c) for case RC206's delivery route planning).

(b) According to the results of VTD and VN, in the same case type, the longer the total vehicle travel distance, the more vehicles are used.

(c) The maximum runtime of the algorithm is 264.94 seconds, and the minimum runtime is 240.57 seconds, indicating that the algorithm can solve cases of different customer distribution types in a relatively short time, providing satisfactory solutions for decision-makers, demonstrating feasibility and effectiveness.

The delivery route planning for cases C109, R209, and RC206 is shown in Figure 3. In Figure 3, the black diamond represents the warehouse, the black dots represent customers, and solid lines represent vehicle travel routes. From Figure 3, it can be observed that the C-type case has customers located at a certain distance from the warehouse, with customers concentrated in specific areas. Most nearby customers in each region are served by the same vehicle. In contrast, the R and RC-type cases have customers distributed randomly, and vehicles first serve customers nearest to the warehouse before proceeding to more distant customers.

C. Comparative Analysis of Open and Closed Vehicle Routing Strategies

Experiments were done comparing the open and closed vehicle routing strategies (where vehicles return to the warehouse after completing deliveries) of the IACA algorithm utilizing a variety of case types to assess the efficacy of the open vehicle strategy. The results are presented in Table II, where the symbols have the same meanings as those in Table I.

Based on the experimental results in Table II-III:

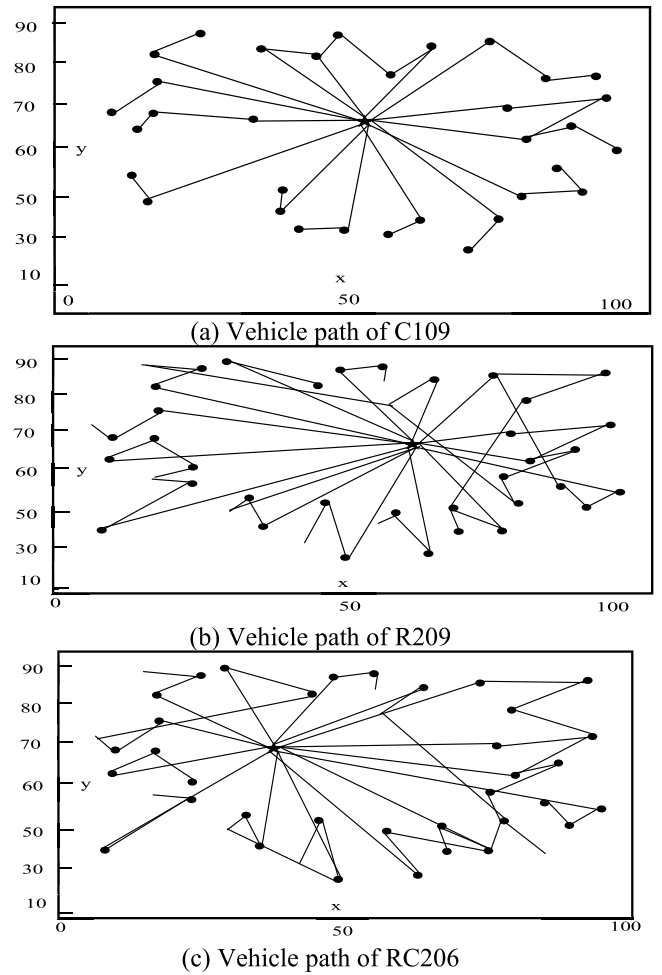


Fig. 3. Delivery route planning for cases C109, R209, and RC206.

TABLE II
CALCULATION RESULTS OF OPEN VEHICLE ROUTING STRATEGY

IN	Open vehicle routing strategy				
	TC/dollar	VUT/min	VTD/km	VN/vehicle	RT/s
C104	2927.9	3090.5	907.52	11	251.79
C105	5334.9	6447.3	1192.4	18	276.3
C106	4624.6	5413.2	1173.6	16	262.81
R206	3214.5	3379.3	1131.6	12	268.28
R207	2734.1	3111.4	1127.2	10	256.04
R208	1900.8	2333.6	946.13	7	245.27
RC203	3489.3	4295.2	1466.5	12	259.35
RC204	2150.6	2499.7	899.94	8	246.57
RC205	4591.3	5302	1479.6	16	275.37

(a) In all cases, the total delivery cost (TC) obtained by the open vehicle routing strategy is significantly lower than that obtained by the closed vehicle routing strategy.

(b) The total vehicle usage time (VUT) obtained by the open vehicle routing strategy is considerably lower than that obtained by the closed vehicle routing strategy.

TABLE III
CALCULATION RESULTS OF CLOSED VEHICLE ROUTING STRATEGY

IN	Closed vehicle routing strategy				
	TC/dollar	VUT/min	VTD/km	VN/vehicle	RT/s
C104	3256.4	3519.2	1389.7	12	301.29
C105	5784.6	7279.5	1953.1	19	300.86
C106	4795.5	5982.7	1763.7	16	287.77
R206	3506.6	3686.4	1351.2	13	280.64
R207	2974.8	3247	1340.2	11	298.62
R208	2372.5	2572.7	1067.3	9	282.6
RC203	3958.3	4525.5	1893.7	14	282.88
RC204	2457.9	2857.2	1099.3	9	284.56
RC205	5148.1	5824.6	2021.1	18	295.76

TABLE IV
CALCULATION RESULTS OF DIFFERENT CALCULATION STANDARDS FOR VEHICLE USAGE TIME

IN	Calculation based on total vehicle usage time				
	TC/dollar	VUT/min	VTD/km	VN/vehicle	RT/s
C101	6619.1	8061.3	1291	22	276.71
C102	5333.4	6442.4	1468.3	18	270.08
R204	2136.6	2453	1028.8	8	245.97
R205	3607.6	4023	1269.2	13	262.65
RC201	5249.6	6162.9	1469.8	18	273.41
RC202	4627.4	5422.4	1578.6	16	275.11

TABLE V
CALCULATION RESULTS OF DIFFERENT CALCULATION STANDARDS FOR VEHICLE USAGE COSTS

IN	The total distance traveled by the vehicle is used as the calculation standard.				
	TC/dollar	VUT/min	VTD/km	VN/vehicle	RT/s
C101	7612.6	10040	990.29	24	270.37
C102	6934.8	8446.9	95143	23	267.29
R204	3184.8	3947.2	847.94	11	239.46
R205	5366.8	5886.9	1028.3	19	254.63
RC201	6637	8787.6	1359.8	21	273.43
RC202	6876.3	7585.3	1335.7	24	266.97

(c) The total vehicle travel distance (VTD) obtained by the open vehicle routing strategy is significantly lower than that obtained by the closed vehicle routing strategy.

Based on the results in Table IV-V, the following conclusions can be drawn:

(a) Under different cost calculation criteria, the algorithm presented in this paper can find satisfactory solutions for various criteria.

(b) When using the total vehicle usage time (VUT) as the cost calculation criterion, the total delivery cost (TC) for all

cases is significantly lower than when using the total vehicle travel distance (VTD) as the cost criterion. On average, TC is 25.91% lower, with the highest reduction being 32.92% (case R204), and the lowest being 13.05% (case C101).

(c) According to the number of vehicles used (VN), when using VUT as the cost criterion, the number of vehicles used is significantly lower than when using VTD as the cost criterion. This suggests that, under dynamic road network conditions, VUT should be used as the cost criterion for vehicle usage.

D. Comparative Analysis of Different Algorithms

To validate the feasibility and effectiveness of the Improved Ant Colony Algorithm (IACA) proposed in this paper, a Genetic Algorithm (GA) program was also developed for solving the OTDVRP. Under unchanged parameters, IACA was compared with Ant Colony Algorithm (ACA) and GA. ACA was set to use only the information concentration heuristic factor and visibility heuristic factor in the transfer rule, without adaptive pheromone heuristic factors and expected heuristic factors. The consumer electronics industry is the very embodiment of these features of supply chain management and related dangers. Planning, sourcing, production, distribution, and returns are the five most important stages of supply chain management. The duties of a supply chain management include cost control, shortfall prevention, and reduction of supplies. The consumer electronics sector is the perfect example of these supply chain management characteristics and associated risks.

Due to missed revenue in the latter scenario and improperly applied investment in the former, both scenarios are expensive. Inventory risk is significant for the consumer electronics sector since it is influenced by a number of factors, including product value, rate of obsolescence, supply and demand unpredictability, and product variety.

The GA used a population size of 100, mutation probability of 0.1, crossover probability of 0.9, and a maximum of 600 iterations. The results for each case are presented in figure 4(a-c), where the symbols have the same meanings as those in Table I.

Based on the results in figure 4(a-c):

(a) Regarding total delivery cost (TC), IACA outperforms both ACA and GA for all cases. IACA's solutions are on average 9.02% lower than those obtained by ACA, with the highest reduction being 15.94% (case RC102) and the lowest being 5.29% (case R201). Compared to GA, IACA's solutions are on average 8.59% lower, with the highest reduction being 18.01% (case C103) and the lowest being 1.68% (case R201).

(b) In terms of total vehicle usage time (VUT), IACA consistently obtains the best results for all cases, with solutions being on average 7.35% lower than ACA's, the highest reduction being 15.52% (case R201), and the lowest reduction being 3.99% (case R202). When compared to GA, IACA's solutions are on average 16.21% lower, with the highest reduction being 25.19% (case C103) and the lowest being 5.33% (case R201).

(c) IACA achieves significantly better results for the number of vehicles used (VN) across all cases compared to the other two algorithms.

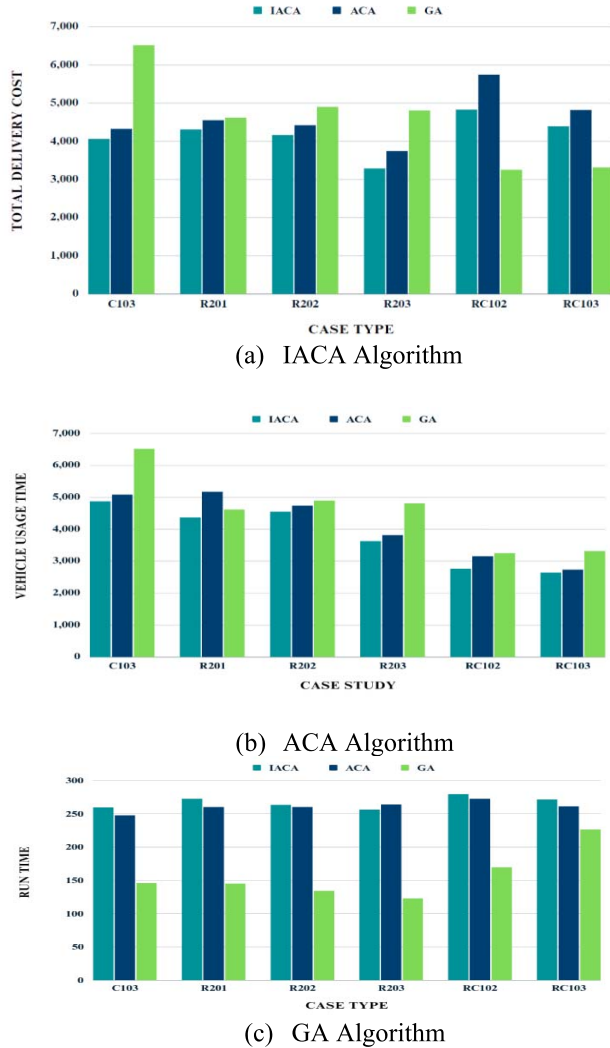


Fig. 4. Calculation results of different algorithms.

VI. CONCLUSION

In this study, a novel method for calculating vehicle travel times was developed, tailored to the dynamic nature of urban road networks. Based on this, an Open Time-Dependent Vehicle Routing Problem (OTDVRP) optimization model was made. It considered customer demand, delivery time windows, product freshness, open-route logistics, and freshness constraints. The primary aim of the model was to minimize total delivery costs. To address this complex problem, an Improved Ant Colony Algorithm (IACA) was introduced. The practicality and effectiveness of the proposed method were rigorously evaluated through simulation experiments, which included a variety of case scenarios. The outcomes of these experiments indicated that adopting open vehicle routing strategies could significantly lower overall delivery costs in e-commerce. This approach saved vehicle usage time and effectively reduced travel distances and the number of vehicles required. One important thing that the study showed was that total vehicle usage time is a crucial cost metric, especially for new e-commerce businesses that need to use dynamic urban road networks. In performance comparisons

between the Improved Ant Colony Algorithm, traditional Ant Colony Algorithms (ACA), and Genetic Algorithms (GA), it was consistently observed that IACA outperformed the other algorithms regarding optimization results. As suggested by the study, future research directions could incorporate factors such as vehicle carbon emissions and fuel consumption. This would contribute to aligning operational efficiencies in fresh e-commerce logistics with environmental sustainability goals, thereby promoting a harmonious balance between efficient logistics and environmental protection.

REFERENCES

- [1] C. De Cauwer, W. Verbeke, J. Van Mierlo, and T. Coosemans, "A model for range estimation and energy-efficient routing of electric vehicles in real-world conditions," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 7, pp. 2787–2800, Jul. 2020, doi: [10.1109/TITS.2019.2918019](https://doi.org/10.1109/TITS.2019.2918019).
- [2] E. Mejri, S. Kelouwani, Y. Dubé, N. Henao, and K. Agbessou, "Energy efficient order picking routing for a pick support automated guided vehicle (Ps-AGV)," *IEEE Access*, vol. 10, pp. 108832–108847, 2022, doi: [10.1109/ACCESS.2022.3212797](https://doi.org/10.1109/ACCESS.2022.3212797).
- [3] Z. Yang, H. Liu, Y. Chen, X. Zhu, Y. Ning, and W. Zhu, "UEE-RPL: A UAV-based energy efficient routing for Internet of Things," *IEEE Trans. Green Commun. Netw.*, vol. 5, no. 3, pp. 1333–1344, Sep. 2021, doi: [10.1109/TGCN.2021.3085897](https://doi.org/10.1109/TGCN.2021.3085897).
- [4] Z. Fang and J. Mao, "Energy-efficient elevating transfer vehicle routing for automated multi-level material handling systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 17, no. 3, pp. 1107–1123, Jul. 2020, doi: [10.1109/TASE.2019.2921631](https://doi.org/10.1109/TASE.2019.2921631).
- [5] M. Shen, G. Luo, and N. Xiao, "EEPC: A framework for energy-efficient parallel control of connected cars," *IEEE Trans. Parallel Distrib. Syst.*, vol. 31, no. 1, pp. 64–79, Jan. 2020, doi: [10.1109/TPDS.2019.2930500](https://doi.org/10.1109/TPDS.2019.2930500).
- [6] K. Hao et al., "An energy-efficient routing void repair method based on an autonomous underwater vehicle for UWSNs," *IEEE Sensors J.*, vol. 21, no. 4, pp. 5502–5511, Feb. 2021, doi: [10.1109/JSEN.2020.3030019](https://doi.org/10.1109/JSEN.2020.3030019).
- [7] C. Liu, J. Wang, W. Cai, and Y. Zhang, "An energy-efficient dynamic route optimization algorithm for connected and automated vehicles using velocity-space-time networks," *IEEE Access*, vol. 7, pp. 108866–108877, 2019, doi: [10.1109/ACCESS.2019.2933531](https://doi.org/10.1109/ACCESS.2019.2933531).
- [8] M. E. Bruni, S. Khodaparasti, and G. Perboli, "Energy efficient UAV-based last-mile delivery: A tactical-operational model with shared depots and non-linear energy consumption," *IEEE Access*, vol. 11, pp. 18560–18570, 2023, doi: [10.1109/ACCESS.2023.3247501](https://doi.org/10.1109/ACCESS.2023.3247501).
- [9] H. Huang, A. V. Savkin, and C. Huang, "Round trip routing for energy-efficient drone delivery based on a public transportation network," *IEEE Trans. Transp. Electr.*, vol. 6, no. 3, pp. 1368–1376, Sep. 2020, doi: [10.1109/TTE.2020.3011682](https://doi.org/10.1109/TTE.2020.3011682).
- [10] J. Baek, S. I. Han, and Y. Han, "Energy-efficient UAV routing for wireless sensor networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 2, pp. 1741–1750, Feb. 2020, doi: [10.1109/TVT.2019.2959808](https://doi.org/10.1109/TVT.2019.2959808).
- [11] Q. Sang, H. Wu, L. Xing, H. Ma, and P. Xie, "An energy-efficient opportunistic routing protocol based on trajectory prediction for FANETs," in *IEEE Access*, vol. 8, pp. 192009–192020, 2020, doi: [10.1109/ACCESS.2020.3032956](https://doi.org/10.1109/ACCESS.2020.3032956).
- [12] Y.-W. P. Hong, R.-H. Cheng, Y.-C. Hsiao, and J.-P. Sheu, "Power-efficient trajectory adjustment and temporal routing for multi-UAV networks," *IEEE Trans. Green Commun. Netw.*, vol. 4, no. 4, pp. 1106–1119, Dec. 2020, doi: [10.1109/TGCN.2020.3014101](https://doi.org/10.1109/TGCN.2020.3014101).
- [13] L. Berghman, Y. Kergosien, and J. Billaut, "A review on integrated scheduling and outbound vehicle routing problems," *Eur. J. Oper. Res.*, vol. 311, no. 1, pp. 1–23, 2023, doi: [10.1016/j.ejor.2022.12.036](https://doi.org/10.1016/j.ejor.2022.12.036).
- [14] J. L. Vilas-Boas, J. J. P. C. Rodrigues, and A. M. Alberti, "Convergence of distributed ledger technologies with digital twins, IoT, and AI for fresh food logistics: Challenges and opportunities," *J. Ind. Inf. Integr.*, vol. 31, Feb. 2023, Art. no. 100393, doi: [10.1016/j.jii.2022.100393](https://doi.org/10.1016/j.jii.2022.100393).
- [15] A. A. Laghari, A. K. Jumani, R. A. Laghari, and H. Nawaz, "Unmanned aerial vehicles: A review," *Cognitive Robotics*, vol. 3, pp. 8–22, Jan. 2023, doi: [10.1016/j.cogr.2022.12.004](https://doi.org/10.1016/j.cogr.2022.12.004).
- [16] K. Braekers, K. Ramaekers, and I. Van Nieuwenhuysse, "The vehicle routing problem: State of the art classification and review," *Comput. Ind. Eng.*, vol. 99, pp. 300–313, Sep. 2016,

- [17] Y. Jiang, B. Bian, and Y. Liu, "Integrated multi-item packaging and vehicle routing with split delivery problem for fresh agri-product emergency supply at large-scale epidemic disease context," *J. Traffic Transp. Eng. (English Edition)*, vol. 8, no. 2, pp. 196–208, 2021.
- [18] A. Hussain, S. U. Khan, N. Khan, M. Shabaz, and S. W. Baik, "AI-driven behavior biometrics framework for robust human activity recognition in surveillance systems," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107218.
- [19] A. Hussain, Z. A. Khan, T. Hussain, F. U. M. Ullah, S. Rho, and S. W. Baik, "A hybrid deep learning-based network for photovoltaic power forecasting," *Complexity*, vol. 2022, Oct. 2022, Art. no. 7040601.
- [20] A. Hussain, S. U. Khan, N. Khan, I. Rida, M. Alharbi, and S. W. Baik, "Low-light aware framework for human activity recognition via optimized dual stream parallel network," *Alex. Eng. J.*, vol. 74, pp. 569–583, Jul. 2023.
- [21] E. Pourrahmani and M. Jaller, "Crowdshipping in last mile deliveries: Operational challenges and research opportunities," *Socio-Econ. Plan. Sci.*, vol. 78, Dec. 2021, Art. no. 101063, doi: [10.1016/j.seps.2021.101063](https://doi.org/10.1016/j.seps.2021.101063).
- [22] H. Chen, J. Liu, J. Wang, and Y. Xun, "Towards secure intra-vehicle communications in 5G advanced and beyond: Vulnerabilities, attacks and countermeasures," *Veh. Commun.*, vol. 39, Feb. 2023, Art. no. 100548, doi: [10.1016/j.vehcom.2022.100548](https://doi.org/10.1016/j.vehcom.2022.100548).
- [23] J. R. Villalobos, W. E. Soto-Silva, M. C. González-Araya, and R. G. González-Ramírez, "Research directions in technology development to support real-time decisions of fresh produce logistics: A review and research agenda," *Comput. Electron. Agric.*, vol. 167, Dec. 2019, Art. no. 105092, doi: [10.1016/j.compag.2019.105092](https://doi.org/10.1016/j.compag.2019.105092).
- [24] J. Xu, Z. Yang, Z. Wang, J. Li, and X. Zhang, "Flexible sensing enabled packaging performance optimization system (FS-PPOS) for lamb loss reduction control in E-commerce supply chain," *Food Control*, vol. 145, Mar. 2023, Art. no. 109394, doi: [10.1016/j.foodcont.2022.109394](https://doi.org/10.1016/j.foodcont.2022.109394).
- [25] J. Ren, H. Li, M. Zhang, and C. Wu, "Massive-scale graph mining for E-commerce cold chain analysis and optimization," *Future Gener. Comput. Syst.*, vol. 125, pp. 526–531, Dec. 2021, doi: [10.1016/j.future.2021.06.057](https://doi.org/10.1016/j.future.2021.06.057).
- [26] D. Zhang, Z. Shen, and Y. Li, "Requirement analysis and service optimization of multiple category fresh products in online retailing using importance-Kano analysis," *J. Retail. Consum. Serv.*, vol. 72, May 2023, Art. no. 103253, doi: [10.1016/j.jretconser.2022.103253](https://doi.org/10.1016/j.jretconser.2022.103253).
- [27] Y. Wang, L. Lei, D. Zhang, and L. H. Lee, "Towards delivery-as-a-service: Effective neighborhood search strategies for integrated delivery optimization of E-commerce and static O2O parcels," *Transp. Res. Part B, Methodol.*, vol. 139, pp. 38–63, Sep. 2020, doi: [10.1016/j.trb.2020.06.003](https://doi.org/10.1016/j.trb.2020.06.003).



Siraj Zahran received the Doctor of Philosophy degree in mechanical and industrial engineering - supply chain management from Ryerson University. He is current working as an Assistant Professor with the Industrial Engineering Department, University of Business and Technology, Jeddah, Saudi Arabia. His area of research includes supply chain management, consignment stock, delay-in-payments, and business development.